

INTRO TO ITWS — QUIZ 1 MASTER CRIB SHEET

Format: Plain-text + Markdown Design Goal: Maximum information density, fast lookup, exam-aligned explanations Chronological Order: By lecture / lab progression Keywords: Bolded All vocab defined

LECTURE 1 — INTRO TO ITWS & WEB SCIENCE

Core Concepts

ITWS (Information Technology & Web Science) studies how technology, people, and information systems interact at scale.

The course is not just coding — it includes networks, web architecture, social impact, economics, and governance.

The web is a socio-technical system: technical infrastructure + human behavior.

Key Vocabulary (Defined)

Web — A global system of interlinked hypertext documents accessed via the Internet using HTTP/HTTPS.

Internet — A global network of networks that communicate using the TCP/IP protocol suite.

Socio-technical system — A system involving both technical components and human/social components.

Scalability — The ability of a system to handle growth in users, data, or traffic without failure.

Emergent behavior — System-level outcomes that arise from many individual interactions (e.g., viral content).

Web Science

Web Science — An interdisciplinary field that studies:

How the web works

How it evolves

How it affects society

How society affects it

The Web Science Method

Observe a web-based phenomenon

Form a hypothesis

Collect data

Analyze results

Intervene or propose solutions

Evaluate outcomes

Exam angle: used to analyze social problems, misinformation, network effects.

How This Appears on the Quiz

Q: What is Web Science and why does it matter?

Answer: Web Science is an interdisciplinary field that studies the web as a socio-technical system, combining technology, human behavior, economics, and governance. It matters because the web influences communication, commerce, politics, and social behavior at global scale, and understanding it allows us to analyze, predict, and improve how web-based systems affect society.

Q: How is ITWS different from a programming-only course?

Answer: ITWS differs from a programming-only course because it focuses not just on writing code, but on understanding how information technology, networks, and web systems interact with people, organizations, and society. The course emphasizes architecture, protocols, social impact, and system-level thinking rather than only implementation.

Q: Explain how the web is a socio-technical system.

Answer: The web is a socio-technical system because it consists of technical infrastructure such as servers, protocols, and code, combined with human users whose behavior shapes how the system evolves. Social factors like trust, incentives, and norms influence how the technology is used, while technical design choices affect social outcomes.

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LECTURE 2 — NETWORKS, PROTOCOLS, TCP/IP

Core Concepts

Communication over the internet happens in layers.

Each layer has a specific responsibility.

Data is wrapped as it moves down the stack and unwrapped as it moves up.

What is a Network?

A network transfers information between devices.

A switch connects devices within a network.

A router connects different networks.

Key Vocabulary (Defined)

Protocol — A formal set of rules governing how data is formatted, transmitted, and received.

Packet — A formatted unit of data transmitted over a network.

Encapsulation — Wrapping data with protocol-specific headers at each layer.

Latency — Delay in data transmission.

Bandwidth — Maximum data transfer capacity.

TCP/IP Model (4 Layers)

Application → Transport → Internet → Link

HTTP → TCP/UDP → IP → Ethernet/WiFi

TCP Three-Way Handshake

SYN — Client requests connection

SYN-ACK — Server acknowledges and agrees

ACK — Client confirms

PURPOSE: Ensures both sides are ready before data transfer.

HTTP Request Cycle

DNS lookup (usually UDP)

TCP handshake

HTTP GET request

HTTP response

Connection maintained or closed

Key Vocabulary (Networking)

DNS — Domain Name System; maps domain names to IP addresses.

IP Address — Numerical identifier for a device on a network.

IPv4 — 32-bit addressing (~4.3 billion addresses).

IPv6 — 128-bit addressing (~340 undecillion addresses).

MAC Address — Hardware-level identifier for network interfaces.

How This Appears on the Quiz

Q: How does your computer know where to go when you type a URL?

Answer: When a URL is entered, the computer first performs a DNS lookup to translate the domain name into an IP address. Once the IP address is known, the computer establishes a TCP connection with the destination server and then uses HTTP to request the web resource.

Q: Why is the TCP three-way handshake required?

Answer: The TCP three-way handshake is required to establish a reliable connection between the client and server. It ensures that both sides are ready to send and receive data, agree on connection parameters, and confirms that communication is possible in both directions before data transmission begins.

Q: Explain the role of DNS in web browsing.

Answer: DNS maps human-readable domain names to numerical IP addresses that computers use for routing data. Without DNS, users would need to remember IP addresses instead of domain names, making the web impractical for human use.

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LECTURE 3 — HTML, XHTML, HTML5

Core Concepts

HTML is semantic markup, not just presentation.

Tags describe meaning, not appearance.

Browsers interpret structure; CSS controls styling.

HTML Versions

HTML 4.01 — Older, presentation-heavy

XHTML 1.0 — HTML rewritten as strict XML

HTML5 — Modern, semantic, extensible, “living standard”

Key Vocabulary (Defined)

Markup language — A language that annotates text with tags to describe structure and meaning.

Semantic HTML — HTML that conveys meaning (e.g., <header>, <nav>, <article>).

DOCTYPE — Tells the browser what type/version of document to expect.

Metadata — Data that describes data.

Basic HTML5 Structure (Annotated)

```
<!-- PURPOSE: Declares the document type as HTML5 -->
<!DOCTYPE html>

<!-- PURPOSE: Root element of the HTML document -->
<html>

<!-- PURPOSE: Contains metadata, styles, scripts, and title -->
<head>
<title>Page Title</title> <!-- PURPOSE: Appears in browser tab -->
</head>

<!-- PURPOSE: Contains all visible page content -->
<body>
<p>Hello World</p> <!-- PURPOSE: Displays a paragraph -->
</body>
</html>
```

Block vs Inline Elements

Block elements — Take full width, create line breaks (<p>, <div>, <h1>)

Inline elements — Flow within text (<a>, ,)

How This Appears on the Quiz

Q: What is the difference between HTML and XHTML?

Answer: HTML is a forgiving markup language that allows flexibility in syntax, while XHTML is a stricter version of HTML written using XML rules. XHTML requires proper nesting, lowercase tags, quoted attributes, and all elements to be closed, whereas HTML allows browsers to correct errors automatically.

Q: What is the fundamental advantage of HTML5?

Answer: The fundamental advantage of HTML5 is that it provides semantic elements, native multimedia support, and extensibility without requiring external plugins. This allows for more meaningful document structure, better accessibility, and improved support for modern web applications.

Q: Why is semantic markup important?

Answer: Semantic markup is important because it conveys the meaning of content to machines such as search engines, screen readers, and other software. This improves accessibility, search engine optimization, and the ability of systems to interpret and process web content correctly.

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LECTURE 4 — CSS & SELECTORS

Core Concepts

CSS controls presentation, not content.

Multiple rules may apply → cascade & specificity decide winner.

Three Ways to Use CSS

External stylesheet (BEST)

Embedded <style> in <head>

Inline style="" (avoid)

CSS Rule Structure

```
/* PURPOSE: Selects paragraph elements */
```

```
p {
```

```
color: red; /* PURPOSE: Sets text color */
```

```
}
```

Selectors & Specificity (High Exam Value)

Specificity order (highest → lowest):

Inline styles

ID selectors (#id)

Class selectors (.class)

Element selectors (p)

Later rules override earlier ones if specificity is equal.

Example Quiz Scenario (Explained)

```
<div id="mainBlock" class="header blockTitle">
```

```
<p>Main Title</p>
```

```
</div>
```

```
p { color: red; } /* element selector */
```

```
#blockTitle { color: blue; } /* ID selector (does NOT match) */
```

```
p#mainBlock.header.blockTitle {
```

```
color: green;
```

```
}
```

Result: Red, because the ID selector does not apply and the compound selector is invalid.

CSS Box Model

Content

Padding

Border

Margin

How This Appears on the Quiz

Q: Which CSS rule wins and why?

Answer: The CSS rule that wins is the one with the highest specificity. If specificity is equal, the rule that appears later in the stylesheet overrides earlier ones. Inline styles have the highest priority, followed by ID selectors, class selectors, and element selectors.

Q: Explain CSS specificity.

Answer: CSS specificity is a ranking system that determines which style rule is applied when multiple rules target the same element. Specificity is calculated based on the type of selectors used, with IDs outweighing classes, and classes outweighing element selectors.

Q: Why are external stylesheets preferred?

Answer: External stylesheets are preferred because they separate content from presentation, allow styles to be reused across multiple pages, improve maintainability, and enable browsers to cache stylesheets for better performance.

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LECTURE 5 — XML & STRUCTURED DATA

Core Concepts

XML describes data, not presentation.

Tags are user-defined.

XML is strict and rule-based.

Key Vocabulary (Defined)

XML — Extensible Markup Language.

Well-formed — Follows XML syntax rules.

DTD — Document Type Definition; defines valid structure.

Schema — XML-based alternative to DTD.

Entity — Encoded reserved character (<, &).

XML Rules (Exam Favorites)

Tags must close

Case-sensitive

Attributes must be quoted

Single root element

Proper nesting

XML Example (Annotated)

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- PURPOSE: XML declaration -->
<book>
<title>Example</title> <!-- PURPOSE: Data element -->
</book>
```

XML vs HTML

How This Appears on the Quiz

Q: Why use XML instead of HTML?

Answer: XML is used instead of HTML when the goal is to store, transport, or describe data rather than display it. XML allows developers to define custom tags and enforces strict structure, making it ideal for data exchange between systems.

Q: What does it mean for XML to be well-formed?

Answer: An XML document is well-formed if it follows all XML syntax rules, including having a single root element, properly nested and closed tags, case-sensitive tag names, and quoted attribute values.

Q: How are XML and XHTML related?

Answer: XHTML is HTML written using XML syntax rules. This means XHTML documents must be well-formed XML documents while still representing web content intended for display in browsers.

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LECTURE 6 — AI: GENERATIVE VS AGENTIC

Core Concepts

AI tools differ in control vs autonomy.

Understanding the difference matters professionally and ethically.

Key Vocabulary (Defined)

Generative AI — Produces outputs in response to prompts (code, text).

Agentic AI — Executes tasks autonomously toward a goal.

Prompt engineering — Crafting effective inputs to guide AI output.

Comparison

CLEAR Prompt Method

Concise

Logical

Explicit

Adaptive

Reflective

How This Appears on the Quiz

Q: Compare generative and agentic AI.

Answer: Generative AI produces responses, such as text or code, based on user prompts and requires the user to execute and evaluate the results. Agentic AI operates more autonomously by taking a goal and performing actions such as reading files, writing code, and iterating on solutions with minimal user intervention.

Q: What are the risks of over-reliance on agentic AI?

Answer: Over-reliance on agentic AI can lead to a lack of understanding of underlying systems, reduced ability to debug or evaluate solutions, and potential ethical or security risks if the AI makes decisions without proper oversight.

Q: Why must developers still understand fundamentals when using AI?

Answer: Developers must understand fundamentals so they can evaluate AI-generated solutions, identify errors, ensure security and correctness, and make informed design decisions. AI amplifies existing skills but cannot replace foundational knowledge.

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LABS — WEBSITE & VM DEPLOYMENT

Core Concepts

Permissions matter

Servers run as users

Authentication controls access

Key Commands (Annotated)

```
sudo chown -R www-data:www-data /var/www/html
```

PURPOSE: Gives Apache ownership so it can read/write files

```
sudo usermod -a -G www-data yourRCSid
```

PURPOSE: Adds user to web server group for permissions

```
sudo nano /var/www/html/iit/.htaccess
```

PURPOSE: Configures directory-level authentication

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“Describe steps to modify site access.”

“Explain why permissions are required.”

“What does Apache need to serve pages?”

CASE DISCUSSION — KUEHCHIC DESSERTS

Key Concepts

Stakeholders — Lim, Tan, Kumar, customers, suppliers.

Triple Constraint — Scope, Time, Cost.

Critical Path — Longest sequence of dependent tasks.

Free Float — Delay allowed without affecting successors.

Total Float — Delay allowed without affecting project end.

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Q: Identify the stakeholders and their influence.

Answer: The primary stakeholders include Lim, Tan, and Kumar, who directly influence project decisions; customers, whose feedback affects product viability; and suppliers, who impact production. The founders have the greatest influence because they control strategy and execution.

Q: Which project constraint is most important and why?

Answer: Time is the most important constraint because the company aims to launch by a specific date. Delays could affect market entry and momentum, making schedule management critical to project success.

Q: What are the major project risks?

Answer: Major risks include task delays on the critical path, limited manpower, insufficient testing of products, and technical challenges related to website development. These risks could prevent the business from launching on schedule.

QUICK LOOKUP INDEX (PANIC SECTION)

TCP handshake → SYN / SYN-ACK / ACK

CSS specificity → Inline > ID > Class > Element

HTML vs XHTML → forgiving vs strict

XML rules → close tags, root element

Web Science → study of web + society

Generative vs Agentic → answer vs action